

MicroServices

Study Guide

Module 6

**Developing BookStore –V.1.4
(Eureka and Load Balancers)**

**Author
Srinivas Dande**





CourseCube®
(Formerly Java Learning Center)



Developing BookStore V-1.4

Service Discovery with Netflix Eureka

- As we know these days, there is a lot of momentum around Microservices.
- The transition from Monolithic to Microservice based architecture gives many benefits for future in terms of **maintainability, scalability, high availability** etc.
- There are many challenges also while doing this migration.
- One of them is to maintain individual Microservices addresses.
- This task can be hugely complex – depending on number of services and their dynamic nature. If whole infrastructure is distributed and there is some replication as well, then maintaining this service addresses becomes harder.
- To solve this, There is a concept called '**Service registration and discovery**' where one dedicated server is responsible to maintain the registry of all the Microservice that has been deployed and removed.
- We can use Eureka as **Service Registry and Discovery Server**.
- Eureka Comes in two parts:
 - a. Eureka server
 - b. Eureka client
- Eureka server for building the service registry server
- Eureka clients which will register themselves and discover other services to call REST APIs.
- When a client microservice registers with Eureka it provides metadata such as host, port, and health indicator thus allowing for other microservices to discover it.
- The discovery server expects a regular heartbeat message from each microservice instance. If an instance begins to consistently fail to send a heartbeat, the discovery server will remove the instance from his registry. This way we will have a very stable ecosystem of Microservices collaborating among each other.



Setup the Eureka Server:

<https://start.spring.io/> --- Spring Boot initializer portal

- 1) **Create a Spring boot project from Spring Boot initializer portal with following**
 - a) **Project** : Maven
 - b) **Language** : Java
 - c) **Spring Boot** : 2.5.6
 - d) **Project Metadata**
 - Group** : com.jlcindia.myeurekaserver
 - Artifact** : MyEurekaServer
 - Name** : MyEurekaServer
 - Description** : Eureka - Service Registry and Discovery Server - Srinivas Dande
 - Package name** : com.jlcindia.myeurekaserver
 - Packaging** : jar
 - Java** : 8
 - e) **Dependencies**
 - Eureka server
 - Spring Boot Actuator

and Click on Generate to download the project.

- 2) **Unzip and import the project into Eclipse as existing maven project.**
- 3) **Delete src/test/java**
- 4) **Update the pom.xml**
- 5) **Now open MyEurekaServerApplication class and add @EnableEurekaServer annotation on the class.**

```
@SpringBootApplication
@EnableEurekaServer
public class MyEurekaServerApplication {

}
```

- 6) **Update application.properties**
spring.application.name=MyEurekaServer
server.port=8761

```
#telling the server not to register himself in the service registry
eureka.client.register-with-eureka=false
eureka.client.fetchRegistry= false
```



7) Start Eureka Server

8) Test Eureka Server

Open browser and go to <http://localhost:8761/>

MyEureka Server: Files required

1. MyEurekaServerApplication.java	2. application.properties
3. pom.xml	

1. MyEurekaServerApplication.java

```
package com.jlcindia.myeurekaserver;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.cloud.netflix.eureka.server.EnableEurekaServer;

@SpringBootApplication
@EnableEurekaServer
public class MyEurekaServerApplication {

    public static void main(String[] args) {
        SpringApplication.run(MyEurekaServerApplication.class, args);
    }

}
```

2. application.properties

```
spring.application.name=MyEurekaServer
server.port=8761

#telling the server not to register himself in the service registry
eureka.client.register-with-eureka=false
eureka.client.fetchRegistry= false
```

3. pom.xml

```
<project ...>
<groupId>com.jlcindia.myeurekaserver</groupId>
<artifactId>MyEurekaServer</artifactId>
<version>0.0.1-SNAPSHOT</version>
<name>MyEurekaServer</name>
<description>Eureka - Service Registry and Discovery Server</description>
```



```
<properties>
<java.version>1.8</java.version>
<spring-cloud.version>2020.0.4</spring-cloud.version>
</properties>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.6</version>
</parent>

<dependencies>
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-actuator</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-netflix-eureka-server</artifactId>
</dependency>
</dependencies>

<dependencyManagement>
<dependencies>
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-dependencies</artifactId>
<version>${spring-cloud.version}</version>
<type>pom</type>
<scope>import</scope>
</dependency>
</dependencies>
</dependencyManagement>

<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```



Step A : Adding Eureka Client in BookPriceMS –V-1.4

Working Steps

1) Add Eureka Client related dependency to pom.xml of BookPriceMS

```
<properties>
<java.version>1.8</java.version>
<spring-cloud.version>2020.0.4</spring-cloud.version>
</properties>

....

<dependency>
  <groupId>org.springframework.cloud</groupId>
  <artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
</dependency>

.....

<dependencyManagement>
<dependencies>
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-dependencies</artifactId>
<version>${spring-cloud.version}</version>
<type>pom</type>
<scope>import</scope>
</dependency>
</dependencies>
</dependencyManagement>
```

2) Add the @EnableEurekaClient annotation on Spring boot application class .

```
@SpringBootApplication
@EnableSwagger2
@EnableEurekaClient
public class BookPriceApplication{

}
```



3) Update application.properties

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
```

4) Start 2 Instances of BookPriceMS

5) Check – 2 Instances of BookPriceMS will be Registered with Eureka Server

Step B : Adding Eureka Client in BookSearchMS – V-1.4

Working Steps

1) Add Eureka Client related dependency to pom.xml of BookSearchMS

```
<dependency>
  <groupId>org.springframework.cloud</groupId>
  <artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
</dependency>
```

2) Add the @EnableEurekaClient annotation on Spring boot application class .

```
@SpringBootApplication
@EnableSwagger2
@EnableFeignClients
@EnableEurekaClient
public class BookSearchApplication {

}
```

3) Update application.properties

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
```

4) Update BookPriceProxy

```
@FeignClient("MyBookPriceMS")
public interface BookPriceProxy {

}
```

5) Start 2 Instances of BookSearchMS

6) Check – 2 Instances of BookSearchMS will be Registered with Eureka Server



Step C : Adding Eureka Client in PlaceOrderMS-V-1.4

Working Steps

1) Add Eureka Client related dependency to pom.xml of PlaceOrderMS

```
<properties>
<java.version>1.8</java.version>
<spring-cloud.version>2020.0.4</spring-cloud.version>
</properties>
...
<dependency>
    <groupId>org.springframework.cloud</groupId>
    <artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
</dependency>
.....
<dependencyManagement>
<dependencies>
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-dependencies</artifactId>
<version>${spring-cloud.version}</version>
<type>pom</type>
<scope>import</scope>
</dependency>
</dependencies>
</dependencyManagement>
```

2) Add the @EnableEurekaClient annotation on Spring boot Main Class.

```
@SpringBootApplication
@EnableSwagger2
@EnableEurekaClient
public class PlaceOrderApplication {

}
```

3) Update application.properties

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
```

4) Start One Instance of PlaceOrderMS

5) Check – One Instance of PlaceOrderMS will be Registered with Eureka Server



Step D : Adding Eureka Client in UserRatingMS-V-1.4

Working Steps

1) Add Eureka Client related dependency to pom.xml of UserRatingMS

```
<properties>
<java.version>1.8</java.version>
<spring-cloud.version>2020.0.4</spring-cloud.version>
</properties>
...
<dependency>
  <groupId>org.springframework.cloud</groupId>
  <artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
</dependency>
.....
<dependencyManagement>
<dependencies>
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-dependencies</artifactId>
<version>${spring-cloud.version}</version>
<type>pom</type>
<scope>import</scope>
</dependency>
</dependencies>
</dependencyManagement>
```

2) Add the @EnableEurekaClient annotation on Spring boot Main Class.

```
@SpringBootApplication
@EnableSwagger2
@EnableEurekaClient
public class UserRatingApplication {

}
```

3) Update application.properties

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
```

4) Start One Instance of UserRatingMS

5) Check – One Instance of UserRatingMS will be Registered with Eureka Server



Step E : Adding Eureka Client in BookStoreWeb -V-1.4

Working Steps

1) Add Eureka Client related dependency to pom.xml of BookStoreWeb

```
<dependency>
  <groupId>org.springframework.cloud</groupId>
  <artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
</dependency>
```

2) Add the @EnableEurekaClient annotation on Spring boot Main class.

```
@SpringBootApplication
@EnableFeignClients
@EnableEurekaClient
public class BookStoreWebApplication{

}
```

3) Update application.properties

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
```

4) Update BookPriceProxy

```
@FeignClient("MyBookPriceMS")
public interface BookPriceProxy {

}
```

5) Update BookSearchProxy

```
@FeignClient("MyBookSearchMS")
public interface BookSearchProxy {

}
```

6) Update PlaceOrderProxy

```
@FeignClient("MyPlaceOrderMS")
public interface PlaceOrderProxy {

}
```



7) Update UserRatingProxy

```
@FeignClient("MyUserRatingMS")
public interface UserRatingProxy {

}
```

8) Start One Instance of BookStoreWeb

9) Check – One Instance of BookStoreWeb will be Registered with Eureka Server

BookPriceMS – V-1.4: Files required

1. BookPriceController.java	Same as V.1.3
2. BookPriceService.java	Same as V.1.3
3. BookPriceServiceImpl.java	Same as V.1.3
4. BookPriceDAO.java	Same as V.1.3
5. BookPrice.java	Same as V.1.3
6. JLCBookPriceConfig.java	Same as V.1.3
7. BookPriceApplication.java	Updated in V1.4
8. application.properties	Updated in V1.4
9. pom.xml	Updated in V1.4

7. BookPriceApplication.java

```
package com.jlcindia.bookprice;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.cloud.netflix.eureka.EnableEurekaClient;
import springfox.documentation.swagger2.annotations.EnableSwagger2;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
@EnableSwagger2
@EnableEurekaClient
public class BookPriceApplication {
    public static void main(String args[]) {
        SpringApplication.run(BookPriceApplication.class, args);
    }
}
```



8. application.properties

spring.application.name=**MyBookPriceMS**

server.port=**9000**

eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true

spring.boot.admin.client.url=http://localhost:5555

management.endpoints.web.exposure.include=*
management.endpoint.health.show-details=always

spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/jlcbookpricedb
spring.datasource.username=root
spring.datasource.password=srinivas

spring.datasource.hikari.connectionTimeout=20000
spring.datasource.hikari.maximumPoolSize=5

logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n

spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update

INFO ENDPOINT CONFIGURATION
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@
info.app.developer=Srinivas Dande
info.app.company=Java Learning Center



9. pom.xml

```
<project ...>

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.bookprice</groupId>
<artifactId>BookPriceMS</artifactId>
<version>1.4</version>
<packaging>jar</packaging>

<name>MyBookPriceMS</name>
<url>https://www.jlcindia.com</url>
<description>This is BookPrice MS developed by Srinivas Dande</description>

<properties>
<java.version>1.8</java.version>
<spring-cloud.version>2020.0.4</spring-cloud.version>
</properties>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.6</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
</dependency>

<dependency>
<groupId>de.codecentric</groupId>
<artifactId>spring-boot-admin-starter-client</artifactId>
<version>2.4.3</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-actuator</artifactId>
</dependency>
```



```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.6</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger2</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger-ui</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<dependencyManagement>
<dependencies>
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-dependencies</artifactId>
<version>${spring-cloud.version}</version>
<type>pom</type>
<scope>import</scope>
</dependency>
```




```
</dependencies>
</dependencyManagement>

<build>
<finalName>MyBookPriceMS</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>

</project>
```

BookSearchMS – V-1.4: Files required

1. BookController.java	Same as V1.3
2. BookPriceInfo.java	Same as V1.3
3. BookInfo.java	Same as V1.3
4. BookInventoryInfo.java	Same as V1.3
5. BookRatingInfo.java	Same as V1.3
6. BookPriceProxy.java	Updated in V1.4
7. BookService.java	Same as V1.3
8. BookServiceImpl.java	Same as V1.3
9. BookDAO.java	Same as V1.3
10. BookRatingDAO.java	Same as V1.3
11. BookInventoryDAO.java	Same as V1.3
12. Book.java	Same as V1.3
13. BookRating.java	Same as V1.3
14. BookInventory.java	Same as V1.3
15. JLCBookSearchConfig.java	Same as V1.3
16. BookSearchApplication.java	Updated in V1.4
17. application.properties	Updated in V1.4
18. pom.xml	Updated in V1.4



6.BookPriceProxy.java

```
package com.jlcindia.booksearch;

import org.springframework.cloud.openfeign.FeignClient;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@FeignClient("MyBookPriceMS")
public interface BookPriceProxy {

    @GetMapping("/bookPrice/{bookId}")
    public BookPriceInfo getBookPrice(@PathVariable Integer bookId);

    @GetMapping("/offeredPrice/{bookId}")
    public double getOfferedPrice(@PathVariable Integer bookId);
}
```

16. BookSearchApplication.java

```
package com.jlcindia.booksearch;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.cloud.netflix.eureka.EnableEurekaClient;
import org.springframework.cloud.openfeign.EnableFeignClients;
import springfox.documentation.swagger2.annotations.EnableSwagger2;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
@EnableSwagger2
@EnableFeignClients
@EnableEurekaClient
public class BookSearchApplication {
    public static void main(String args[]) {
        SpringApplication.run(BookSearchApplication.class, args);
    }
}
```



17. application.properties

spring.application.name=MyBookSearchMS

server.port=8000

eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true

spring.rabbitmq.host=localhost
spring.rabbitmq.port=5672
spring.rabbitmq.username=guest
spring.rabbitmq.password=guest

spring.boot.admin.client.url=http://localhost:5555

management.endpoints.web.exposure.include=*
management.endpoint.health.show-details=always

spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/jlcbooksdb
spring.datasource.username=root
spring.datasource.password=srinivas

spring.datasource.hikari.connectionTimeout=20000
spring.datasource.hikari.maximumPoolSize=5

logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n

spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update

INFO ENDPOINT CONFIGURATION
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@
info.app.developer=Srinivas Dande
info.app.company=Java Learning Center



18. pom.xml

```
<project ...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.booksearch</groupId>
<artifactId>BookSearchMS</artifactId>
<version>1.4</version>
<packaging>jar</packaging>
<name>MyBookSearchMS</name>
<url>https://www.jlcindia.com</url>
<description>This is BookSeach MS developed by Srinivas Dande</description>

<properties>
<java.version>1.8</java.version>
<spring-cloud.version>2020.0.4</spring-cloud.version>
</properties>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.6</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-openfeign</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-amqp</artifactId>
</dependency>

<dependency>
<groupId>de.codecentric</groupId>
<artifactId>spring-boot-admin-starter-client</artifactId>
<version>2.4.3</version>
</dependency>
```



```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-actuator</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.6</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger2</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger-ui</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<dependencyManagement>
<dependencies>
<dependency>
<groupId>org.springframework.cloud</groupId>
```



```
<artifactId>spring-cloud-dependencies</artifactId>
<version>${spring-cloud.version}</version>
<type>pom</type>
<scope>import</scope>
</dependency>
</dependencies>
</dependencyManagement>

<build>
<finalName>MyBookSearchMS</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```

PlaceOrderMS – V-1.4: Files required

1. OrderController.java	Same as V1.3
2. OrderFullInfo.java	Same as V1.3
3. OrderInfo.java	Same as V1.3
4. OrderItemInfo.java	Same as V1.3
5. BookInventoryInfo.java	Same as V1.3
6. OrderService.java	Same as V1.3
7. OrderServiceImpl.java	Same as V1.3
8. OrderDAO.java	Same as V1.3
9. OrderItemDAO.java	Same as V1.3
10. BookInventoryDAO.java	Same as V1.3
11. Order.java	Same as V1.3
12. OrderItem.java	Same as V1.3
13. BookInventory.java	Same as V1.3
14. JLCPlaceOrderConfig.java	Same as V1.3
15.PlaceOrderApplication.java	Updated in V1.4
16.application.properties	Updated in V1.4
17.pom.xml	Updated in V1.4



15.PlaceOrderApplication.java

```
package com.jlcindia.placeorder;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.cloud.netflix.eureka.EnableEurekaClient;
import springfox.documentation.swagger2.annotations.EnableSwagger2;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
@EnableSwagger2
@EnableEurekaClient
public class PlaceOrderApplication {
    public static void main(String args[]) {
        SpringApplication.run(PlaceOrderApplication.class, args);
    }
}
```

16. application.properties

```
spring.application.name=MyPlaceOrderMS

server.port=7000

eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true

spring.boot.admin.client.url=http://localhost:5555

spring.rabbitmq.host=localhost
spring.rabbitmq.port=5672
spring.rabbitmq.username=guest
spring.rabbitmq.password=guest

management.endpoints.web.exposure.include=*
management.endpoint.health.show-details=always

spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/jlcordersdb
spring.datasource.username=root
spring.datasource.password=srinivas
```



```
spring.datasource.hikari.connectionTimeout=20000
spring.datasource.hikari.maximumPoolSize=5

logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n

spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update

# INFO ENDPOINT CONFIGURATION
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@
info.app.developer=Srinivas Dande
info.app.company=Java Learning Center
```

17. pom.xml

```
<project ...>

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.placeorder</groupId>
<artifactId>PlaceOrderMS</artifactId>
<version>1.4</version>
<packaging>jar</packaging>
<name>MyPlaceOrderMS</name>
<url>https://www.jlcindia.com</url>
<description>This is PlaceOrderMS developed by Srinivas Dande</description>

<properties>
<java.version>1.8</java.version>
<spring-cloud.version>2020.0.4</spring-cloud.version>
</properties>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.6</version>
</parent>
```




```
<dependencies>
```

```
<dependency>
```

```
<groupId>org.springframework.cloud</groupId>
```

```
<artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
```

```
</dependency>
```

```
<dependency>
```

```
<groupId>org.springframework.boot</groupId>
```

```
<artifactId>spring-boot-starter-amqp</artifactId>
```

```
</dependency>
```

```
<dependency>
```

```
<groupId>de.codecentric</groupId>
```

```
<artifactId>spring-boot-admin-starter-client</artifactId>
```

```
<version>2.4.3</version>
```

```
</dependency>
```

```
<dependency>
```

```
<groupId>org.springframework.boot</groupId>
```

```
<artifactId>spring-boot-starter-actuator</artifactId>
```

```
</dependency>
```

```
<dependency>
```

```
<groupId>org.springframework.boot</groupId>
```

```
<artifactId>spring-boot-starter-web</artifactId>
```

```
</dependency>
```

```
<dependency>
```

```
<groupId>org.springframework.boot</groupId>
```

```
<artifactId>spring-boot-starter-data-jpa</artifactId>
```

```
</dependency>
```

```
<dependency>
```

```
<groupId>mysql</groupId>
```

```
<artifactId>mysql-connector-java</artifactId>
```

```
<version>5.1.6</version>
```

```
</dependency>
```

```
<dependency>
```

```
<groupId>io.springfox</groupId>
```

```
<artifactId>springfox-swagger2</artifactId>
```

```
<version>2.4.0</version>
```

```
</dependency>
```




```
<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger-ui</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<dependencyManagement>
<dependencies>
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-dependencies</artifactId>
<version>${spring-cloud.version}</version>
<type>pom</type>
<scope>import</scope>
</dependency>
</dependencies>
</dependencyManagement>

<build>
<finalName>MyPlaceOrderMS</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>

</project>
```



UserRatingMS – V-1.4: Files required

1. RatingController.java	Same as V1.3
2. UserRatingInfo.java	Same as V1.3
3. BookRatingInfo.java	Same as V1.3
4. RatingService.java	Same as V1.3
5. RatingServiceImpl.java	Same as V1.3
6. UserRatingDAO.java	Same as V1.3
7. BookRatingDAO.java	Same as V1.3
8. UserRating.java	Same as V1.3
9. BookRating.java	Same as V1.3
10. JLCUserRatingConfig.java	Same as V1.3
11.UserRatingApplication.java	Updated in V1.4
12.application.properties	Updated in V1.4
13.pom.xml	Updated in V1.4

11.UserRatingApplication.java

```
package com.jlcindia.userratings;
```

```
import org.springframework.boot.SpringApplication;  
import org.springframework.boot.autoconfigure.SpringBootApplication;  
import org.springframework.cloud.netflix.eureka.EnableEurekaClient;
```

```
import springfox.documentation.swagger2.annotations.EnableSwagger2;  
/*  
* @Author : Srinivas Dande  
* @company : Java Learning Center  
**/  
  
@SpringBootApplication  
@EnableSwagger2  
@EnableEurekaClient  
public class UserRatingApplication {  
    public static void main(String args[]) {  
        SpringApplication.run(UserRatingApplication.class, args);  
    }  
}
```



12. application.properties

spring.application.name=MyUserRatingMS

server.port=6500

eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true

spring.boot.admin.client.url=http://localhost:5555

spring.rabbitmq.host=localhost
spring.rabbitmq.port=5672
spring.rabbitmq.username=guest
spring.rabbitmq.password=guest

management.endpoints.web.exposure.include=*
management.endpoint.health.show-details=always

spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/jlcratingsdb
spring.datasource.username=root
spring.datasource.password=srinivas

spring.datasource.hikari.connectionTimeout=20000
spring.datasource.hikari.maximumPoolSize=5

logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n

spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update

INFO ENDPOINT CONFIGURATION
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@
info.app.developer=Srinivas Dande
info.app.company=Java Learning Center



13. pom.xml

```
<project ...>

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.userrating</groupId>
<artifactId>UserRatingMS</artifactId>
<version>1.4</version>
<packaging>jar</packaging>
<name>MyUserRatingMS</name>
<url>https://www.jlcindia.com</url>
<description>This is UserRatingMS developed by Srinivas Dande</description>

<properties>
<java.version>1.8</java.version>
<spring-cloud.version>2020.0.4</spring-cloud.version>
</properties>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-amqp</artifactId>
</dependency>

<dependency>
<groupId>de.codecentric</groupId>
<artifactId>spring-boot-admin-starter-client</artifactId>
<version>2.4.3</version>
</dependency>
```



```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-actuator</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.6</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger2</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger-ui</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>
```



```
<dependencyManagement>
<dependencies>
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-dependencies</artifactId>
<version>${spring-cloud.version}</version>
<type>pom</type>
<scope>import</scope>
</dependency>
</dependencies>
</dependencyManagement>

<build>
<finalName>MyUserRatingMS</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>

</project>
```

BookStoreWeb – V-1.4: Files required

1. index.jsp	Same as V1.3
2. myheader.jsp	Same as V1.3
3. myfooter.jsp	Same as V1.3
4. showBooksList.jsp	Same as V1.3
5. showBookInfo.jsp	Same as V1.3
6. showCart.jsp	Same as V1.3
7. orderSuccess.jsp	Same as V1.3
8. addRating.jsp	Same as V1.3
9. ratingSuccess.jsp	Same as V1.3
10. ratingsList.jsp	Same as V1.3
11. bookstore.css	Same as V1.3
12. Book.java	Same as V1.3
13. BookInfo.java	Same as V1.3
14. Order.java	Same as V1.3



15. OrderItem.java	Same as V1.3
16. UserRating.java	Same as V1.3
17. BookStoreController.java	Same as V1.3
18. OrderFullInfo.java	Same as V1.3
19. OrderInfo.java	Same as V1.3
20. OrderItemInfo.java	Same as V1.3
21. UserRatingInfo.java	Same as V1.3
22. BookPriceInfo.java	Same as V1.3
23. BookPriceProxy.java	Updated in V1.4
24. BookSeacrhProxy.java	Updated in V1.4
25. PlaceOrderProxy.java	Updated in V1.4
26. UserRatingProxy.java	Updated in V1.4
27. BookStoreService.java	Same as V1.3
28. BookStoreServiceImpl.java	Same as V1.3
29. JLCBookStoreWebConfig.java	Same as V1.3
30. BookStoreWebApplication.java	Updated in V1.4
31. application.properties	Updated in V1.4
32. pom.xml	Updated in V1.4

23. BookPriceProxy.java

```
package com.jlcindia.bookstoreweb;
```

```
import org.springframework.cloud.openfeign.FeignClient;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@FeignClient("MyBookPriceMS")
public interface BookPriceProxy {

    @GetMapping("/bookPrice/{bookId}")
    public BookPriceInfo getBookPrice(@PathVariable Integer bookId);

    @GetMapping("/offeredPrice/{bookId}")
    public double getOfferedPrice(@PathVariable Integer bookId);
}
```



24. BookSearchProxy.java

```
package com.jlcindia.bookstoreweb;

import java.util.List;

import org.springframework.cloud.openfeign.FeignClient;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@FeignClient("MyBookSearchMS")
public interface BookSearchProxy {

    @GetMapping("/mybooks/{author}/{category}")
    public List<Book> getBooks(@PathVariable String author, @PathVariable String category);

    @GetMapping("/mybook/{bookId}")
    public BookInfo getBookById(@PathVariable Integer bookId);
}
```

25. PlaceOrderProxy.java

```
package com.jlcindia.bookstoreweb;

import java.util.List;

import org.springframework.cloud.openfeign.FeignClient;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@FeignClient("MyPlaceOrderMS")
public interface PlaceOrderProxy {

    @GetMapping("/myorders/{userId}")
    public List<Order> getOrdersByUserId(@PathVariable String userId);

}
```




26. UserRatingProxy.java

```
package com.jlcindia.bookstoreweb;

import java.util.List;

import org.springframework.cloud.openfeign.FeignClient;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@FeignClient("MyUserRatingMS")
public interface UserRatingProxy {

    @GetMapping("/userRatings/{userId}")
    public List<UserRating> getUserRatingByUserId(@PathVariable String userId) ;

}
```

30. BookStoreWebApplication.java

```
package com.jlcindia.bookstoreweb;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.cloud.netflix.eureka.EnableEurekaClient;
import org.springframework.cloud.openfeign.EnableFeignClients;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
@EnableFeignClients
@EnableEurekaClient
public class BookStoreWebApplication {
    public static void main(String args[]) {
        SpringApplication.run(BookStoreWebApplication.class, args);
    }
}
```



31. application.properties

```
spring.application.name=MyBookStoreWeb

server.port=5000

eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true

spring.boot.admin.client.url=http://localhost:5555

spring.rabbitmq.host=localhost
spring.rabbitmq.port=5672
spring.rabbitmq.username=guest
spring.rabbitmq.password=guest

management.endpoints.web.exposure.include=*
management.endpoint.health.show-details=always

logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n

# INFO ENDPOINT CONFIGURATION
info.app.name=@project.name@
info.app.description=@project.description@
info.app.version=@project.version@
info.app.encoding=@project.build.sourceEncoding@
info.app.java.version=@java.version@
info.app.developer=Srinivas Dande
info.app.company=Java Learning Center
```

32.pom.xml

```
<project .....">

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.bookstoreweb</groupId>
<artifactId>BookStoreWeb</artifactId>
<version>1.4</version>
<packaging>jar</packaging>

<name>MyBookStoreWeb</name>
<url>https://www.jlcindia.com</url>
<description>This is BookStoreWeb developed by Srinivas Dande</description>
```



```
<properties>
<java.version>1.8</java.version>
<b>spring-cloud.version>2020.0.4</spring-cloud.version>
</properties>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<b>version>2.5.6</version>
</parent>

<dependencies>

<b>dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-openfeign</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-amqp</artifactId>
</dependency>

<dependency>
<groupId>de.codecentric</groupId>
<artifactId>spring-boot-admin-starter-client</artifactId>
<version>2.4.3</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-actuator</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>
```



```
<dependency>
<groupId>org.apache.tomcat.embed</groupId>
<artifactId>tomcat-embed-jasper</artifactId>
</dependency>

<dependency>
<groupId>jstl</groupId>
<artifactId>jstl</artifactId>
<version>1.2</version>
</dependency>

<dependency>
<groupId>>taglibs</groupId>
<artifactId>standard</artifactId>
<version>1.1.2</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>bootstrap</artifactId>
<version>4.3.1</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>jquery</artifactId>
<version>3.4.1</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<dependencyManagement>
<dependencies>
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-dependencies</artifactId>
<version>${spring-cloud.version}</version>
<type>pom</type>
<scope>import</scope>
```



```
</dependency>
</dependencies>
</dependencyManagement>

<build>
<finalName>MyBookStoreWeb</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>

</project>
```